

Integrating Water Distribution Energy Flexibility in Power Systems Operation

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Abstract—Water distribution systems (WDSs) are energy intensive infrastructures that rely on water storage plants and variable speed pumps for delivering water to the end consumers. The WDSs represent suitable candidates for enhancing the flexibility of power systems operation through optimizing the operation of flexible components (pumps and tanks). However, WDSs have traditionally been treated as infrastructure uncoupled from power system operators' efforts to minimize the operation cost of supplying the electricity demand. In this paper, we propose a model for integrating the energy flexibility of WDSs in power systems operation. More specifically, we consider the water distribution system operators (W-DSOs) as an energy savvy entity, who forecast the water demand as well as the electricity prices of the next day, and run the proposed day-ahead WDS operation model to optimize the operation of pumps and tanks for minimizing the operation cost of their local WDSs, while respecting the hydraulic operating constraints of the WDSs. The W-DSOs, then, send the optimized energy consumption to the power system operator, who incorporates the energy consumption of WDSs in the day-ahead operation of the power system using the network-constrained unit commitment (UC) model. The proposed model is implemented on a 15-node WDS that is powered by a 6-bus power system and the results are presented to study the impacts of optimizing the WDS energy flexibility on power system operations as well as on the W-DSOs.

Index Terms—Water distribution system operators, energy flexibility, power systems operation, water-energy nexus.

I. INTRODUCTION

Water distribution systems (WDSs), water treatment plants (WTPs), and wastewater treatment plants (WWTPs) account for about 3% to 4% of the total electricity used in the United States [1]. Approximately 80% of this total electricity consumption is used for pumping and distributing water and wastewater with the remaining 20% to account for treatment [2]. Further, drinking water and wastewater utilities are typically the largest energy consumers of municipal governments, accounting between 30% to 40% of total energy consumed [3]. As population grows and regulations become more stringent, electricity demand associated with water supply and treatment is projected to increase.

Water and power systems have been traditionally analyzed as two separate uncoupled infrastructure systems. However, in reality when these systems are viewed through the lens of energy interdependence, they are reciprocally and mutually linked. Meeting energy needs requires water, often in large quantities, for mining, fuel production, hydropower, and power plant cooling, and energy is needed for pumping, treatment,

and distribution of water and for collection, treatment, and discharge of wastewater [3]. Therefore, in today's economies, energy and water are tightly linked. The availability of adequate water supplies has a profound impact on the availability of energy, while energy production and power generation activities affect the availability and quality of water [4]. This interrelationship is often referred to as the water-energy nexus.

Predominantly, the water-energy nexus has been studied with respect to the policy and regulatory challenges and its interactions with the food supply, climate change, economic growth, and water rights [5], [6]. Recent work has been done in investigating the impact of climate change on the operation of the hydroplants and reservoir management [7]. From a modeling viewpoint, the water-energy nexus has been studied using different approaches. In [8], the coupling between the power and water is studied in terms of the co-production facilities (e.g., hydroelectric and thermal desalination plants). The cost the power generation, water, and co-production facilities were minimized together in [8] considering plants' physical characteristics and assuming that all the facilities were connected into a single node, ignoring the impacts of transmission and distribution. Additional work has also investigated the role of storage facilities in the optimal dispatch of generating units [9]. In [10], the line capacity limits of a test power system and the hydraulic pipe constraints of a hypothetical 8-node WDS were introduced, leading to a dc optimal power flow problem. The desalination units were used to link the water and power systems. Robust optimization has recently been applied to hydrothermal scheduling problem, however, without modeling the constraints of the power system networks [11]. The same power system modeling omission is widely observed in optimization problems whose primary scope is to schedule the operation of the pumps in water system facilities [12].

In a different approach, the water-energy nexus has been modeled as a physics-based integrated engineering system. In [13] a set of algebraic equations was derived to describe the input-output energy relations between the electricity, water and wastewater systems using the systems modeling language. These models, when combined, can relate the energy and municipal water consumption of a region to the required water withdrawals in an input-output model.

A joint effort between power systems operators and the water and wastewater facilities they serve could create a great opportunity for reducing peak electricity demand and increase the energy efficiency of the energy water and wastewater industry [14]. These efforts are mainly focused on passive energy efficiency and price-based load management strategies. Water system facilities are energy intensive infrastructures and

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therefore suitable candidates for energy efficiency applications. From the power system point of view, the energy efficiency programs could improve the system operations, reduce operation costs, and possibly reduce emission from the electricity sector [15]. Several pilot energy efficiency programs that include loads from water facilities are conducted and their economic impacts on power systems operation is studied [16].

Although there has been a great deal of research in regards to investigating and modeling the water-energy nexus, less attention has been given to account for the dynamic energy flexibility that water system facilities can introduce to the operation of power systems. The water facility owners and the water distribution system operators (W-DSOs) have a natural incentive to act energy savvy and minimize their energy costs. However, the W-DSOs may be concerned about the negative impacts of modifying the facilities operation on the reliable delivery of water to the consumers.

We argue that the technical literature lacks a comprehensive model to analyze the energy flexibility of WDSs considering the underlying physics of their hydraulic constraints and the dynamic operation of their energy intensive components. The modeling approaches in the technical literature do not address the issue of adjusting the operation of water facilities to accommodate water demand fluctuations and to operate the facilities taking into account the geographic topology of the reservoirs in WDSs. The majority of the literature models the water-energy nexus either by introducing desalination facilities as the coupling components between the water and power networks or via using meta-architecture physics based, energy input-output engineering systems. However, co-production facilities such as hydro plants or desalination plants may not always be present in WDSs. Therefore, the coupling point between the water and power system may be absent. In addition, the majority of the dispatch models in literature treat the water-energy nexus at supply side as a single node neglecting the power and water transmission or distribution networks that they feed. This representation neither accounts for the power flow limits of the transmission network nor for the hydraulic constraints in WDSs. Therefore, the dispatch decisions of these models may not be supported by the constrained power or water network characteristics, especially when considering that one of the pumps main role is to overcome water pipe losses.

This paper investigates the water-energy nexus in the context of WDSs, and elaborates WDSs modeling as part of integrating the flexibility of their energy-intensive components (e.g., pumps) and storage plants (tanks) in power systems operation. More specifically, in this paper we develop an operation model for W-DSOs that optimizes the hourly schedules of the WDS flexible components for minimizing the WDSs operating costs. In the proposed optimization model, the WDS energy and cost savings are becoming apparent by re-scheduling the operation of the network's storage and pump facilities, while providing a reliable supply of water demand at all times. The optimized energy consumption of the WDSs is then incorporated in the day-ahead power systems operation, through the solution of the network-constrained unit commitment problem.

The rest of the paper is organized as follows. Section II presents the proposed model for integrating the energy

flexibility of WDSs in power systems operation, while the proposed W-DSO operation model is formulated in Section III. The simulations results on a 6-bus power system and a 15-node WDS are presented in Section IV, and the conclusions are given in Section V.

II. WATER DISTRIBUTION ENERGY FLEXIBILITY IN DAY-AHEAD POWER SYSTEMS OPERATION

The goal of day-ahead power systems operation is to schedule the most economic set of generating units to supply the projected demand of the next day, respecting the operational constraints of the units and power flow limits of the transmission network. Water distribution systems are energy intensive infrastructures that are usually equipped with water storage facilities and variable speed pumps (VSPs) to compensate for water load variations. However, the water tanks and pumps could also provide energy flexibility by coordinating the operation of these flexible components with power systems operation. The energy consumption coordination is also beneficial for the water distribution system operators (W-DSOs) where the overall electricity demand and variable operational cost of the WDSs can be reduced significantly, without affecting the reliable supply of water demand.

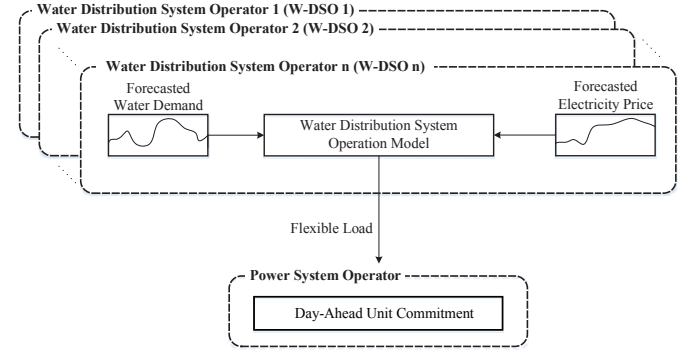


Fig. 1: Proposed water-energy Nexus framework

In this paper, we propose a model to integrate the energy flexibility of WDSs in the day-ahead operation of power systems. The structure of the proposed model is shown in Fig. 1. In Fig. 1, the W-DSOs forecast the water demand as well as the electricity prices of the next day, and run a day-ahead WDS operation model to optimize the operation of pumps and tanks for minimizing the operation cost of their local WDSs, while respecting the hydraulic operating constraints of the WDSs. The WDSs then send the optimized energy consumption to the power system operator, who incorporates the optimized energy consumption of WDSs in the day-ahead power system operation using network-constrained unit commitment (UC) model. Although WDSs loads are usually connected to the power systems at medium voltage grids, their energy consumption flexibility would impact the transmission network operation. Here we do not intend to repeat the well-understood UC equations, and refer the readers to our past works for detailed mixed-integer linear programming formulation of the networked-constrained UC formulation incorporating demand flexibility [17]–[21]. In the next Section III, we present the formulation of proposed W-DSO model.

III. WATER DISTRIBUTION SYSTEM OPERATION

In this section, we present the proposed model to optimize the operation of WDSs. The proposed W-DSO operation optimization problem models the components of the water distribution network (including pipes, water pumps, water tanks, reservoirs and water loads) and its associated hydraulic constraints in detail. From a mathematical point of view, a WDS is represented by a directed graph $G = (\mathcal{N}, \mathcal{P})$, where $\mathcal{N}$ is the set of nodes representing reservoirs, tanks, and load points, and $\mathcal{P}$ is the set of arcs comprising pipes and pumps. The pressure head of the nodes is represented by the decision variable $N \times 1$ vector $\mathbf{H}_t$, while the volumetric flow rate of pipes and pumps is shown by the decision variable $P \times 1$ vector $\mathbf{Q}_t$, where N and P represent the total number of nodes and arcs respectively. Water flow through the network is induced by different pressure levels at the nodes, shown by $\mathbf{A}^T \mathbf{H}_t$. The matrix $\mathbf{A}$ is the $N \times P$ node-arc incidence matrix, where rows and columns correspond to nodes and arcs of the network respectively. Generally, the flow rate in a pipe can be maintained either through sufficient elevation difference between two nodes or by producing a pressure head by pumping. For example, if a WDS is fed via a ground water supply system so that gravity does not suffice, then a pumping station is required to raise certain pressure node potentials.

The operation costs of WDSs include the variable costs incurred by purchasing water fed into the network and the energy needed to operate the pumps. The objective function of the proposed WDS operation model is to supply the forecasted water demand at minimum operation cost, formulated as:

$$\min \sum_{t \in T} (\mathbf{K}^T \Theta \mathbf{W}_t^R + \mathbf{E}_t^T \mathbf{1}^T \pi_{t,b}) \quad (1)$$

where $\mathbf{W}_t^R$ is the $N \times 1$ vector of water inflow at nodes; Θ is the $R \times N$ reservoirs-nodes incidence matrix that maps the water reservoirs to nodes, and R is the number of reservoirs; $\mathbf{K}$ is the $R \times 1$ reservoirs flat water price; $\pi_{t,b}$ is the forecasted electricity price at power transmission bus b where the WDS is supplied from; $\mathbf{E}_t$ is the $U \times 1$ decision variable vector of pumps' electricity consumption at time t , where U is the total number of pumps. The objective function (1) is constrained to the WDSs water flow and hydraulic constraints (2)-(9). The flow continuity law (2) states that the sum of inflows and outflows is equal to zero for each node, where $\mathbf{W}_t^D$ is the $N \times 1$ vector of water demand at nodes, and $\mathbf{C}_t$ and $\mathbf{D}_t$ are respectively the $T \times 1$ decision variables vectors of water inflow and outflow of tanks, where T is the total number of tanks. $\mathbf{\Lambda}$ is the $N \times T$ nodes-tanks incidence matrix where rows and columns correspond to nodes and tanks of the network respectively. The water inflow and outflow of tanks are explicitly modeled in (2), so the W-DSO may utilize the water storage capacity of tanks to store water and serve water demand in future times. The dynamic operation of tanks is governed by the tank state equation (3), where $\mathbf{V}_t$ is the $T \times 1$ water volume decision variable vector of tanks at time t . $\mathbf{V}_t$ depends on the pressure head decision variable vector, $\mathbf{H}_t$, and the surface of the tanks $\mathbf{\Delta}$ in (4). Note that $\mathbf{\Delta}$ is the $T \times T$ diagonal matrix that contains the surface parameters for each

tank. The tanks' water capacity constraints are imposed in (5), where $\underline{\mathbf{V}}$ and $\overline{\mathbf{V}}$ are the minimum and maximum capacity limits of the tanks. The maximum inflow/outflow rate limits are also enforced in (6) and (7), where $\overline{\mathbf{M}}$ represents the flow limit. Constraint (8) sets the water volume stored in tanks at time $t = 1$. The volumetric flow rate decision variable vector of pipes and pumps $\mathbf{Q}_t$ is bounded in (9), where the limit $\overline{\mathbf{Q}}$ is based on the pipe diameter and construction material.

$$\mathbf{W}_t^R - \mathbf{W}_t^D = \mathbf{A} \mathbf{Q}_t + \mathbf{\Lambda} [\mathbf{C}_t - \mathbf{D}_t] \quad \forall t \quad (2)$$

$$\mathbf{V}_{t+1} = \mathbf{V}_t + \mathbf{C}_t - \mathbf{D}_t \quad \forall t \quad (3)$$

$$\mathbf{V}_t = \mathbf{\Delta} \mathbf{\Lambda}^T \mathbf{H}_t \quad \forall t \quad (4)$$

$$\underline{\mathbf{V}} \leq \mathbf{V}_t \leq \overline{\mathbf{V}} \quad \forall t \quad (5)$$

$$\mathbf{C}_t \leq \overline{\mathbf{M}} \quad \forall t \quad (6)$$

$$\mathbf{D}_t \leq \overline{\mathbf{M}} \quad \forall t \quad (7)$$

$$\mathbf{V}_0 = \mathbf{V}^{init} \quad (8)$$

$$-\overline{\mathbf{Q}} \leq \mathbf{Q}_t \leq \overline{\mathbf{Q}} \quad \forall t \quad (9)$$

The pressure head (potential) decision variables, $\mathbf{H}_t$, are modified by pressure loss in pipes, and are controlled by pressure increase in the pumps. The pressure loss along a pipe due to friction is described by the empirical *Hazen-Williams* or head loss constraint (10), where $\mathbf{G}$ is the $P \times P$ diagonal matrix that contains the resistance parameters for each pipe, and $\Pi(\cdot)$ is the unit sign function. The nodal pressure heads $\mathbf{H}_t$ are constrained in (11) in order to guarantee minimum and maximum pressure levels at the nodes. These bounds are usually determined based on the WDS design specifications (e.g., population served, types of dwellings and firefighting requirements). The nodal pressure heads are fixed at the reservoir nodes in (12), reflecting the fact that water at source nodes is not pressurized and it exploits a fixed geographical height, where $\mathbf{Z}$ is the $R \times 1$ vector that contains the elevation parameters of the reservoir nodes.

$$\mathbf{A}^T \mathbf{H}_t = \mathbf{G} \Pi(\mathbf{Q}_t) |\mathbf{Q}_t|^{1.852} \quad \forall t \quad (10)$$

$$\underline{\mathbf{H}} \leq \mathbf{H}_t \leq \overline{\mathbf{H}} \quad \forall t \quad (11)$$

$$\Theta \mathbf{H}_t = \mathbf{Z} \quad \forall t \quad (12)$$

The pressure head increase produced by VSPs is given in (13), where ω_t is the $U \times 1$ speed decision variable vector of pumps and $\mathbf{B}$ is the $U \times P$ pump-arc incidence matrix, where rows and columns correspond to pumps and arcs of the network respectively. The power consumption of pumps is given in (14), where a, b, c, d are coefficients associated with the performance of the pumps. The VSPs allow manipulation of the relation between potential raise and volumetric flow rate. Positive flow rate in pumps is guaranteed in (15). Note that $\odot$ in (13)-(14) denotes componentwise matrix multiplication.

$$\mathbf{B} \mathbf{A}^T \mathbf{H}_t = \omega_t^2 \odot \left(a - b \left(\frac{1}{\omega_t} \odot \mathbf{B} \mathbf{Q}_t \right)^c \right) \quad \forall t \quad (13)$$

$$\mathbf{E}_t = \omega_t^3 \odot \left(d - e \left(\frac{1}{\omega_t} \odot \mathbf{B} \mathbf{Q}_t \right) \right) \quad \forall t \quad (14)$$

$$\mathbf{B} \mathbf{Q}_t \geq 0 \quad \forall t \quad (15)$$

In summary, (1)-(15) represent the proposed operation model for WDSs, which optimizes the operation of tanks and pumps for minimizing the operation cost in (1). The proposed model is a nonlinear optimization problem, with the nonlinearities represented by (10), (13), and (14).

IV. STUDY RESULTS

The proposed model to integrate the flexibility of WDSs on power systems operation is implemented on a test WDS connected to the 6-bus test system. The test WDS, shown in Fig. 2, is based on a simplified version of a real WDS, and includes 15 nodes, 14 pipes, 3 pumps, 2 tanks, and one reservoir representing the only external water source. The data for this WDS is available in [22]. The volume of the tanks equal to 15% of the WDS daily water peak load, i.e., $0.3 \text{ m}^3/\text{sec}$, which is divided equally between the two tanks. The peak energy consumption of each WDS is assumed to be equal to 20% of the peak demand of the respective power system bus. We consider the 6-bus power system [23], shown in Fig. 3, which supplies the energy consumption of three WDSs, connected to the three load buses 2, 4 and 5.

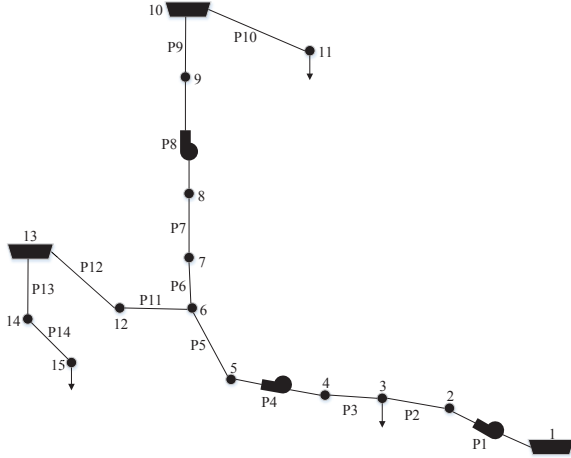


Fig. 2: Schematic representation of the test WDS

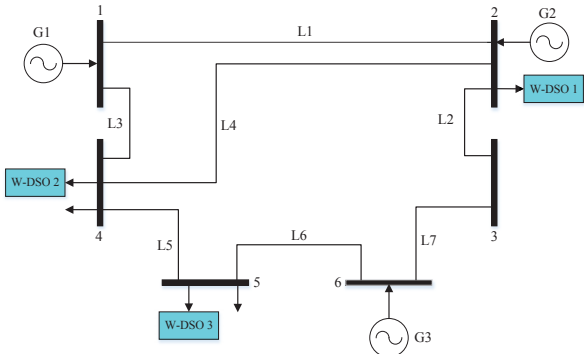


Fig. 3: Online diagram of the 6-bus system

Two cases are studied here. In Case 1, the three WDSs are simulated without optimizing the energy consumption of flexible components (pumps and tanks). In this case, the energy consumption of W-DSOs is not optimized. The resulting operating cost of the 6-bus system in Case 1 is \$80,998.2.

In Case 2, the energy consumption of WDSs are optimized using the model proposed in Section III. In this case, the

WDSs would be considered as flexible loads for the six-bus power system. This load flexibility is achieved by optimizing the operation of pumps and the water inflow and outflow of the tanks, for minimizing the operation cost of the WDSs in (1), while respecting the WDSs operation constraints in (2)-(14). The optimized energy consumption of WDSs are then sent to the power system operator, who runs the day-ahead network-constrained UC problem using the adjusted WDSs loads. The operating cost of the 6-bus system in Case 2 is reduced to \$77,905.4 which shows the operational cost benefits to the power system in the presence of WDSs energy flexibility.

The total electricity consumption of the three WDSs in two cases are shown in Fig. 4. In Fig. 4, the energy consumption of WDSs is shifted to the off peak hours after optimization, while meeting the water demand over the day-ahead scheduling horizon. This has become possible by the availability of water storage capacity in tanks. The proposed model schedules the pumps to store water in the two tanks during hours 1-8 in Fig. 5. The stored water is then released during the peak energy hours 9-19, which has considerably reduced the need for pumps' operation to supply the water demand in the hours. As a result, the energy cost of the three WDSs is considerably reduced in Table I. In Table I, more energy cost savings are achieved for W-DSO 2 and W-DSO 3 than for W-DSO 1, which shows that the energy optimization of WDSs is more attractive, financially, in power system buses 4 and 5. This result also highlights the fact that the energy costs of WDSs are linked to the operational characteristics of power systems.

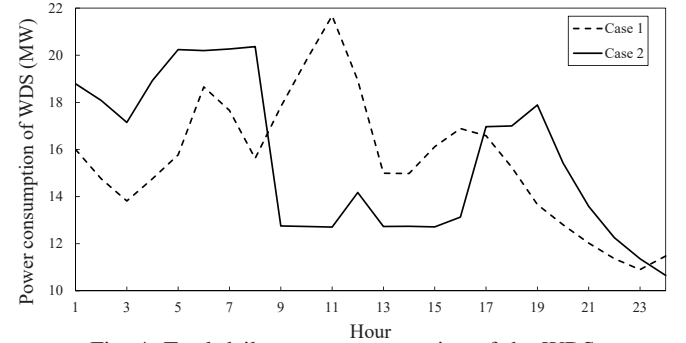


Fig. 4: Total daily power consumption of the WDSs

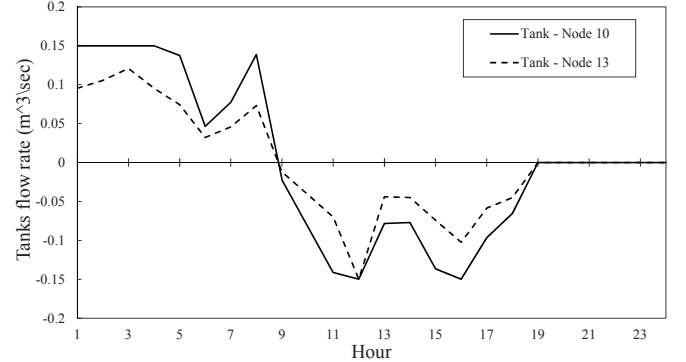


Fig. 5: Optimized flow rate of the tanks in Case 2

The total load of the 6-bus power system in Cases 1 and 2 (before and after optimizing the energy consumption of WDSs) is shown in Fig. 6. In Fig. 6, optimizing the energy consumption of WDSs has smoothed out the daily load

TABLE I: Energy Cost Savings of the WDSs in Case 2

Case	Total Daily Energy Cost (\$)		
	W-DSO 1	W-DSO 2	W-DSO 3
Case 1	6360.2	6967.3	6852.1
Case 2	5800.2	6135.5	6071.9
Energy cost savings in Case 2	559.9	831.8	780.2

profile and has reduced the peak load of the 6-bus system, as compared to Case 1 where the energy consumption of WDSs is not optimized. In Case 1, with the nonsmooth load profile, all the three generating units of the 6-bus system are committed to supply the daily load profile. However, the more expensive unit G2 is not anymore committed in Case 2 where the energy optimization of WDSs smooths out the load profile. The energy optimization of WDSs also modifies the marginal electricity prices in the 6-bus system. In Fig. 7, the average locational marginal price (LMP) curve in Case 1 is non-smooth and shows a sharp spike in hour 9, while the average LMP curve in Case 2 is smooth and represent the benefit of WDSs energy optimization to all the energy customers.

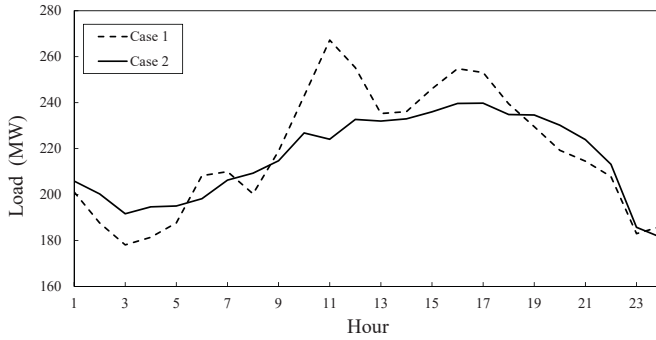


Fig. 6: Total load of the 6-Bus system in Cases 1 and 2

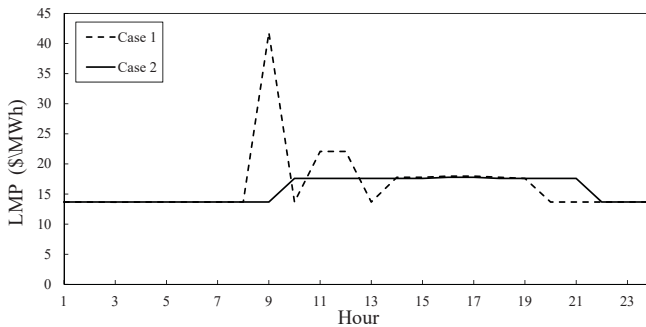


Fig. 7: Average locational marginal electricity prices in the 6-Bus system in Cases 1 and 2

V. CONCLUSION

This paper proposed a model for integrating the energy flexibility of water distribution systems (WDSs) in power systems operation. A energy savvy water distribution system operator (W-DSO) is considered who optimizes the operation of water pumps and water tanks for minimizing its energy costs, while respecting the hydraulic operation constraints of the WDSs. The proposed model is implemented on a test 15-node WDS that is powered by a 6-bus power system.

The simulation results indicate that both the power system operators as well as the W-DSOs benefit from implementing the proposed model. The proposed model would reduce the operation cost of the power system, reduce the peak electricity demand, smooth out the load profile, and reduce the electricity prices. In addition, the W-DSOs enjoy considerable energy cost savings when they optimize their energy consumption.

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